ELectronics EXpress

Measurement and modeling of spectral transmission tilt in WDM systems due to stimulated Raman scattering

Zuyuan He,[1a)] **Chia-Chen Chang,**[2] **Jun Bao,**[2] **Wenxin Zheng,**[2] **Erin Sahinci,**[2] **and Waqar Mahmood**[2]

[1] *Department of Electronic Engineering, University of Tokyo*
7–3–1 Hongo, Bunkyo-ku, Tokyo 113–8656, Japan
[2] *CIENA Corporation*
930 International Drive, Linthicum, Maryland 21090, USA
a) *ka@sagnac.t.u-tokyo.ac.jp*

Abstract: Stimulated Raman scattering (SRS) degrades the performance of wavelength-division-multiplexed (WDM) optical fiber communication systems by amplifying the longer-wavelength channels at the expense of shorter-wavelength channels, and making the spectral profile of transmission tilted. The spectral transmission tilt due to SRS of single-mode fiber span is measured by using a technique to separate it from the total loss profile of the fiber. An empirical model is developed to predict the transmission tilt for single-mode fiber of any fiber length under various optical power levels. The application of the model in test and evaluation of gain-flattened optical fiber amplifiers is introduced.

Keywords: Fiber characterization, optical transmission, stimulated Raman scattering, wavelength division multiplex, optical fiber amplifier
Classification: Photonic devices, circuits, and systems

References

[1] A. R. Chraplyvy, "Optical power limits in multi-channel wavelength division multiplexed systems due to stimulated Raman scattering," *Electron. Lett.*, vol. 20, no. 2, pp. 58–59, 1984.
[2] A. R. Chraplyvy, "Limitation on lightwave communications imposed by optical-fiber non-linearities," *J. Lightwave Technol.*, vol. 8, no. 10, pp. 1548–1557, 1990.
[3] V. J. Mazurczyk, G. Shaulov, and E. A. Golovchenko, "Accumulation of gain tilt in WDM amplified systems due to Raman crosstalk," *IEEE Photon. Technol. Lett.*, vol. 12, no. 11, pp. 1573–1575, 2000.
[4] D. N. Christodoulides and R. B. Jander, "Evolution of stimulated Raman crosstalk in wavelength division multiplexed systems," *IEEE Photon. Technol. Lett.*, vol. 8, no. 12, pp. 1722–1724, 1996.

DOI: 10.1587/elex.1.311
Received August 19, 2004
Accepted August 31, 2004
Published September 10, 2004

IEICE Electronics Express, Vol.1, No.11, 311–316

[5] X. Y. Zou, S.-M. Hwang, and A. E. Willner, "Compensation of Raman scattering and EDFA's nonuniform gain in ultra-long-distance WDM links," *IEEE Photon. Technol. Lett.*, vol. 8, no. 1, pp. 139–141, 1996.
[6] S. Tariq and J. C. Palais, "A computer model of non-dispersion-limited stimulated Raman scattering in optical fiber multiple-channel communications," *J. Lightwave Technol.*, vol. 11, no. 12, pp. 1914–1924, 1993.
[7] A. Tomita, "Cross talk caused by stimulated Raman scattering in single-mode wavelength-division multiplexed systems," *Opt. Lett.*, vol. 8, no. 7, pp. 412–414, 1983.
[8] S. Bigo, S. Gauchard, A. Bertaina, and J.-P. Hamaide, "Experimental investigation of stimulated Raman scattering limitation on WDM transmission over various types of fiber infrastructures," *IEEE Photon. Technol. Lett.*, vol. 11, no. 6, pp. 671–673, 1999.
[9] Corning Incorporated, *Optical Fiber Product Information*, 2001.

1 Introduction

It has been well recognized that stimulated Raman scattering (SRS) is one of the major nonlinear optical phenomena that can degrade the performance of wavelength-division-multiplexed (WDM) optical communication systems, especially for dense WDM systems, where the large number of channels involved lead to high optical power levels in the fiber [1, 2]. In such systems, SRS is known to induce power exchange between channels, which is referred to as SRS crosstalk. In general, longer-wavelength channels are amplified at the expense of shorter-wavelength channels [3, 4]. This effect primarily affects the power distribution of the channels, and results in a tilted spectral profile of the total transmission of the system. In an amplified system, the amplifiers have to be designed to compensate the SRS-induced spectral transmission tilt [5].

The effects of SRS in WDM systems have been addressed previously in several studies. The SRS crosstalk among channels has been treated numerically or analytically [1, 4, 6]. In Ref. [7], the Raman crosstalk was measured in a 21-km-long single-mode fiber, using two CW laser diodes operating at 1.26-μm (the pump channel) and 1.34-μm (the signal channel) wavelengths. Ref. [8] measured the impact of SRS on the power distribution of a 32-channel multiplex after a fixed length of 100-km transmission over various fiber types.

In this paper, we developed a technique to measure the SRS-induced spectral transmission profile, or referred to as Raman gain profile hereafter, of single-mode fiber. This technique separates the Raman gain profile from the total loss profile of the fibers. With this method, Raman gain profiles of different fiber spans with different lengths are measured. Furthermore, a model is developed to predict the Raman gain profile at any fiber length under various optical power levels. The model can be used in evaluation of the SRS-induced penalties in WDM system, or in design, test, and evaluation of gain-flattened erbium-doped fiber amplifier (EDFA). As an example of the application of the model, a "virtual span," i.e., a numerical simulator

DOI: 10.1587/elex.1.311
Received August 19, 2004
Accepted August 31, 2004
Published September 10, 2004

is developed to simulate a physical fiber span, which is used in the test and evaluation of gain-flattened EDFA. Experimental results show good agreements between using a physical fiber span and using the "virtual span".

2 Measurement

As SRS occurs, the total transmission (gain or loss) of an optical fiber span is a function of the optical power propagating in the fiber. It can be represented in decibel as

$$T(P) = A + G(P) \quad \text{(dB)}. \tag{1}$$

A is the background loss of the fiber, including the attenuation of the fiber and insertion loss of connections. A is independent to the optical power, but a function of light wavelength and proportional to the fiber length when ignoring the loss from connections. $G(P)$ is the Raman gain. P is the total optical power in watt of all the channels propagating in the fiber.

If we increase the optical power to $2P$, then the total transmission becomes

$$T(2P) = A + G(2P) \quad \text{(dB)}. \tag{2}$$

By subtracting Eq. (2) from 2 times Eq. (1), we get

$$A = 2T(P) - T(2P) \quad \text{(dB)}, \tag{3}$$

where we used $G(2P) = 2G(P)$ because Raman gain in decibel is proportional to the optical power in watt approximately when the Raman interaction is weak [4, 7]. Then we can obtain

$$G(P) = T(P) - A \quad \text{(dB)}. \tag{4}$$

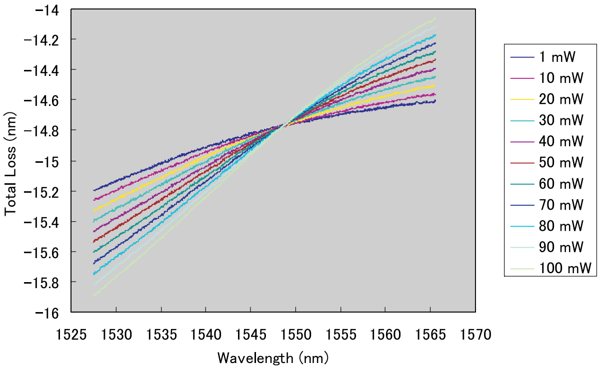
Eqs. (3) and (4) imply that both the Raman gain and the background loss of the optical fiber can be simply extracted from multiple measurements of total insertion loss at different optical power levels.

Employing this technique, we measured the Raman gain profiles of several optical fiber spans of different lengths. The measurement setup is an insertion loss test setup with a flat broadband light source. Assuming each channel has equal optical power in a multi-channel system, we can use a flat broadband source to simulate a multi-channel source of large number of channels. In our experiment, we used a flat amplified spontaneous emission (ASE) source, which is pump-wavelength-stabilized, with maximum output power 22 dBm, and ripple less than 1.39 dB over 1527.5 $\sim$ 1565.5 nm. The input power to the fiber was adjusted with a GPIB-controlled attenuator, whose settings were calibrated together with the flat ASE source before measurement. The optical power spectra were measured with an optical spectrum analyzer at the input and output ends of the fiber to get the total insertion loss of the fiber span. We tested standard single mode fiber (SMF) (Corning, SMF-28) and non-zero dispersion shifted fiber (NZDSF) (Lucent, TrueWave), each of 25 km, 50 km, and 75 km, respectively.

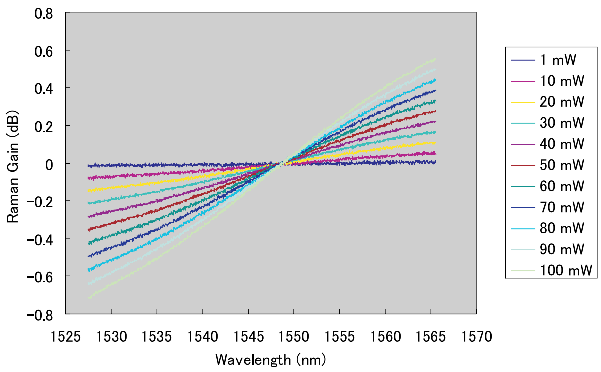
DOI: 10.1587/elex.1.311
Received August 19, 2004
Accepted August 31, 2004
Published September 10, 2004

IEICE Electronics Express, Vol.1, No.11, 311–316

Fig. 1 (a) shows the total insertion loss profile at various input powers for 75-km SMF. By applying Eqs. (3) and (4) on the data of Fig. 1 (a), we get the Raman gain profile at input power levels from 1 to 100 mW as shown in Fig. 1 (b). From Fig. 1 (b), we can see that the longer-wavelength channels get gains at the expenses of losses of shorter-wavelength channels. For 75-km SMF, the total Raman tilt over the whole band can reach about 1.3 dB under 100-mW input power. We examined the relation of SRS-induced tilt to the input power, which exhibits an exact linear function in good agreement with the theory.



(a)



(b)

Fig. 1. (a) Total insertion loss, (b) Raman gain of 75-km SMF.

3 Modeling

In principle, the SRS is proportional to the optical power, and nonlinear to the wavelength [6]. Due to SRS, longer wavelength channels get net gain in the expenses of net loss in shorter wavelength channels. As power increases, Raman gain profile "rotates" around a convergence point in the center of the band. Due to its central symmetrical feature, it can be represented as the 3rd order polynomial of wavelength. Based on experimental investigation data, we develop an empirical expression to simulate the Raman gain profile in single-mode fiber. It is a function of total input power P, wavelength λ,

DOI: 10.1587/elex.1.311
Received August 19, 2004
Accepted August 31, 2004
Published September 10, 2004

IEICE Electronics Express, Vol.1, No.11, 311–316

and fiber length L represented as

$$G = L_{eff}(g_0 + g_1\lambda + g_2\lambda^2 + g_3\lambda^3)P \quad \text{(dB)}. \tag{5}$$

$L_{eff} = [1 - \exp(-\alpha L)]/\alpha$ is the effective length, where α is the attenuation coefficient of the fiber. For SMF-28, $\alpha = 0.04375$/km [9]. Based upon the experimental data, we can determine all the coefficients of $g_0 \sim g_3$.

Fig. 2 shows an example of the model-generated Raman gain profiles compared with the measurement results. Figs. 2 (a) and (b) are for 50-km and 25-km SMF-28 fiber spans, respectively. It is shown that the model can predict the Raman profile in good agreement with measurement results for any fiber length and at various input power levels. We obtained similar results for NZDSF also.

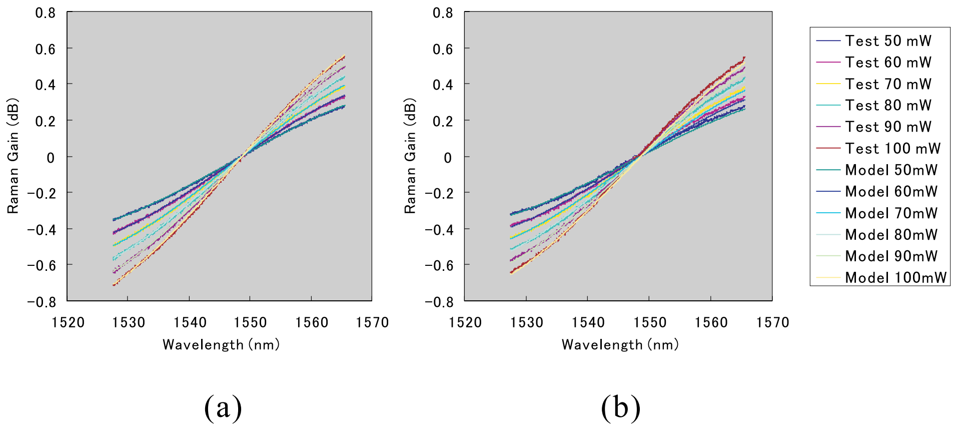


Fig. 2. Comparison of model generated Raman gain profiles to test results for SMF. (a) 50 km, (b) 25 km.

4 Application in test and evaluation of gain-flattened EDFA

One of the biggest problems in gain-flattened EDFA manufacturing, once it is built correctly, is the variability of the testing. The same amplifier that fails on one test station may pass on another. One of the biggest variations comes from the fiber span variation. It was found per tests that the variation of fiber spans comes easily over the range of the amplifier specification.

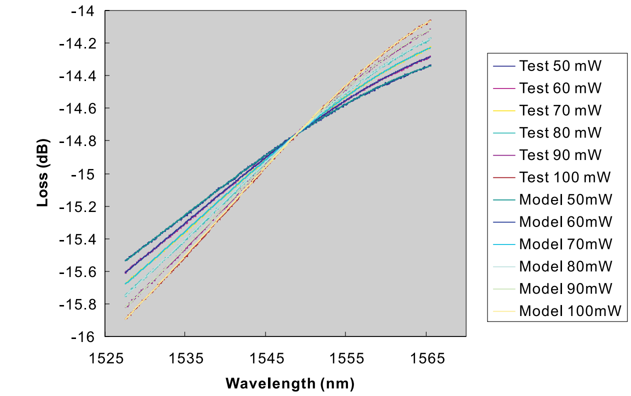
One solution to the problem is using a mathematical "virtual span" to replace the physical fiber span in the test and evaluation. Using this "virtual span" will eliminate the span variation between test stations to ensure the consistency of amplifier test. It also saves the cost of test facilities.

Using the "virtual span" is especially significant for testing the amplifiers in metro communication products, where the fiber spans may have lengths differing from, for example, 40 km to 70 km approximately. It is very hard to install multiple fiber spans of different lengths in each test station.

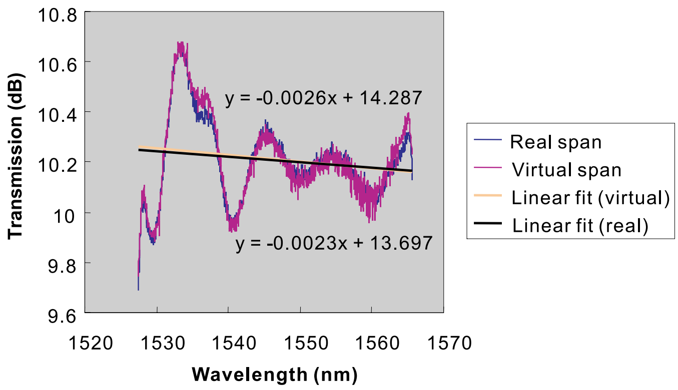
The "virtual span" can be generated by using the Raman gain profile model developed above plus the background loss profile. It can emulate a

DOI: 10.1587/elex.1.311
Received August 19, 2004
Accepted August 31, 2004
Published September 10, 2004

IEICE Electronics Express, Vol.1, No.11, 311–316



(a)



(b)

Fig. 3. (a) Comparison between the "virtual span" and measured physical span of 75-km SMF, (b) using the "virtual span" in test of gain-flattened amplifier.

physical span of different lengths and under various input power. Fig. 3 (a) shows the comparison between the model-generated span and measured physical span of 75-km SMF. We also developed similar emulator for NZDSF.

Fig. 3 (b) gives an example of gain flatness EDFA test using the "virtual span" with comparison to using the real span. It is shown that the test result using "virtual span" agreed well with that using real span. It demonstrated the feasibility to replace the physical span in the test and evaluation of gain flattened optical amplifiers with the "virtual span".

5 Conclusion

The spectral profile of SRS-induced transmission tilt, i.e., Raman gain profile, of single-mode fiber span is measured by using a technique to separate the Raman gain profile from the total loss profile of the optical fiber. An empirical model is developed to predict the Raman gain profile at any fiber length under various optical power levels. The model can be used in design, test, and evaluation of gain-flattened EDFA. It can also be employed in evaluation of the SRS-induced performance degradations in WDM systems.

DOI: 10.1587/elex.1.311
Received August 19, 2004
Accepted August 31, 2004
Published September 10, 2004